Understanding tobacco smoke carcinogen NNK and lung tumorigenesis.

The deleterious effects of tumor-promoting tobacco carcinogen, nitrosamine 4-(methylnitrosamino)-1-(3-pyridyl)-1-butanone (NNK, nicotine-derived nitrosamine ketone) have undoubtedly been stipulated. Though many tobacco agents play a part in the development of lung tumors, the potent effects of NNK remain unmatched. It is therefore critical to distinguish the variety of cofactors involved in NNK-mediated pathogenesis, and the unique pathways necessary for successful cellular biotransformation. Current reviews have consistently identified the strengths of NNK and prospective tumor capabilities. Others have delineated specific cellular factors mediating NNK and lung tumors, and have identified metabolic and signaling pathways largely responsible for NNK activation and tumorigenic initiation. Unique to this review is that it summarizes the extensive network of cofactors and cellular mechanisms that promote NNK-specific lung tumorigenesis. As such, it displays a fuller, more comprehensive overview, bringing us one step closer to understanding the fatal consequences of NNK, thus, discovering new avenues that successfully break the cycle of NNK-mediated lung carcinogenesis.